

Statistical Method Reference Guide

Method Comparison Tables by Problem Category

DDSI Decision Design Assistant — Bootcamp Reference

Purpose

This guide provides a reference for selecting the most appropriate statistical methods based on the problem category, constraints, and stakeholder requirements. Use after a problem has been categorized and assessed for feasibility.

Method Selection Criteria (in priority order)

1. **Problem fit** — Does the method actually answer this type of question?
2. **Data requirements** — Does available data meet minimum needs?
3. **Interpretability** — Can the stakeholder understand and trust the output?
4. **Delivery timeline** — Can it be implemented within available time?
5. **Accuracy / power** — Sufficient analytical power?
6. **Output format** — Compatible with how the user will consume results?

Hypothesis Testing

Method	Best When	Data Needs	Interpret.	Timeline
Two-sample t-test	Comparing means of two groups, roughly normal	30+ per group, continuous	High	1-2 days
Paired t-test	Before/after on same subjects	Paired measurements, normal diffs	High	1-2 days
Mann-Whitney U	Two groups, non-normal or ordinal	20+ per group	High	1-2 days
Chi-square test	Proportions or categorical independence	Expected counts ≥ 5	High	1-2 days
Fisher's exact	Small sample categorical	Any sample size	High	1 day
Proportion z-test	Proportion vs. benchmark	30+ obs, binary outcome	High	1 day

Contact for expertise: *Experimentation & Insights Team*

Regression & Correlation Analysis

Method	Best When	Data Needs	Interpret.	Timeline
Simple linear regression	One predictor, linear	50+ obs, continuous X and Y	High	2-3 days
Multiple linear regression	Multiple predictors, isolate effects	10-20 obs per predictor	Med-High	1-2 wks
Logistic regression	Binary outcome (yes/no)	10+ events per predictor	Med-High	1-2 wks
Pearson correlation	Linear relationship strength	Continuous, bivariate normal	High	1 day
Spearman rank	Non-linear monotonic, ordinal	Ordinal or continuous	High	1 day
Polynomial regression	Curved relationship	Sufficient data across curve	Medium	1-2 wks

Contact for expertise: *Customer Analytics Team*

Time Series Analysis & Forecasting

Method	Best When	Data Needs	Interpret.	Timeline
Moving avg / exp. smoothing	Stable patterns, quick turnaround	12+ periods	High	1-3 days
Seasonal decomposition	Trend + seasonal + residual	2+ seasonal cycles	High	2-5 days
ARIMA / SARIMA	Clear seasonal, univariate	24+ periods, stationary	Medium	3-7 days
Prophet	Multiple seasonalities, holidays	12+ months daily/weekly	High	3-7 days
Holt-Winters	Trend + seasonality, simple	2+ seasonal cycles	High	2-5 days

Contact for expertise: Demand Science Team

Estimation & Confidence Intervals

Method	Best When	Data Needs	Interpret.	Timeline
Sample mean + CI	Estimating population mean	30+ obs (CLT) or normal pop	High	1 day
Proportion CI	Estimating population proportion	np and n(1-p) >= 10	High	1 day
Bootstrap CI	Non-standard stats, unknown dist	100+ obs recommended	Medium	2-3 days
Prediction interval	Individual future observations	Model fit, residual dist known	Medium	2-3 days
Tolerance interval	Capturing proportion of population	Normal or large sample	Medium	2-3 days

Contact for expertise: Customer Analytics Team

Distribution Fitting & Descriptive Analytics

Method	Best When	Data Needs	Interpret.	Timeline
Five-number summary + box plots	Quick distribution overview	Any numeric data	High	1 day
Histogram + KDE	Visualizing distribution shape	50+ observations	High	1 day
QQ plots + Shapiro-Wilk	Testing for normality	5-5000 observations	Medium	1 day
MLE distribution fitting	Fitting parametric distribution	Domain knowledge for candidates	Medium	3-5 days
Kolmogorov-Smirnov	Comparing empirical vs. theoretical	Continuous, 50+ obs	Medium	1-2 days

Contact for expertise: *Data Engineering & Monitoring Team*

Comparative Analysis (Multi-Group)

Method	Best When	Data Needs	Interpret.	Timeline
One-way ANOVA	Comparing means, 3+ groups, normal	20+ per group, equal var	High	2-3 days
Tukey's HSD (post-hoc)	Which groups differ after ANOVA	Same as ANOVA	High	2-3 days
Kruskal-Wallis	3+ groups, non-normal	5+ per group	High	2-3 days
Chi-square independence	Categorical distributions across groups	Expected counts ≥ 5	High	1-2 days
Two-way ANOVA	Two factors, interaction effects	Balanced design, 10+ per cell	Medium	3-5 days

Contact for expertise: *Experimentation & Insights Team*

Sampling & Survey Design

Method	Best When	Data Needs	Interpret.	Timeline
Simple random sampling	Homogeneous population	Sampling frame	High	2-3 days
Stratified sampling	Known subgroups differ	Strat. vars, pop proportions	Med-High	3-5 days
Cluster sampling	Natural groupings (locations)	Cluster definitions, ICC est.	Medium	3-5 days
Power analysis / sample size	How many observations needed	Effect size est., power 0.8	High	1-2 days
Systematic sampling	Ordered list, even coverage	Ordered frame, no periodicity	High	1-2 days

Contact for expertise: *Experimentation & Insights Team*

Bayesian & Probabilistic Estimation

Method	Best When	Data Needs	Interpret.	Timeline
Conjugate Bayesian updating	Simple estimation with natural prior	Prior dist., likelihood, data	Medium	3-5 days
Bayesian A/B testing	Probability one option is better	Same as freq. A/B + prior	Medium	1-2 wks
Empirical Bayes	Many similar estimation problems	Multiple related groups	Low-Med	1-2 wks
Bayesian regression	Full posterior of coefficients	Same as freq. regression + priors	Low-Med	2-3 wks
Monte Carlo simulation	Complex probability, uncertainty	Input distributions, model struct	Medium	2-4 wks

Contact for expertise: *Risk Analytics Team*

How to Present Recommendations

When recommending methods, always present:

- **Top 3 ranked methods** with clear rationale for ranking
- **Primary tradeoff** for each (e.g., "faster but less precise")
- **Data requirements** that must be met
- **Estimated delivery timeline**
- **Recommended contact** for deeper expertise
- **Output format compatibility** — connect method to user's interaction model

Example Output Format

"Based on your problem (Regression & Correlation), data availability (3 years daily park data, 8 predictors), timeline (2 weeks), and need for an interpretable executive presentation:

Rank 1: Multiple Linear Regression — Best fit. Interpretable coefficients. Tradeoff: assumes linear relationships. Deliverable in 5-7 days. Contact: Customer Analytics.

Rank 2: Spearman Correlation Matrix — Quick diagnostic first. No linearity assumption. Lower depth but fast. Deliverable in 2-3 days.

Rank 3: Logistic Regression — Better if outcome is binary (satisfied vs. not). Deliverable in 1-2 weeks.

"Recommendation: Start with Spearman correlations as a 2-day diagnostic, then move into Multiple Linear Regression for the full analysis."